

Spatiotemporal reconstruction of groundwater depletion in India's oldest coalfield using machine learning and GIS, with InSAR and fractal analysis insights

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Abstract—Rapid mining expansion, urban growth, and climate variability have intensified groundwater (GW) stress in several industrial regions of India. This is particularly true in the country's oldest coalfield, the Raniganj area. However, long-term high-resolution monitoring of groundwater storage (GWS) and groundwater levels (GWLs) in such heavily modified terrains remains constrained by sparse observations and data gaps. This study develops a data-driven machine-learning framework to reconstruct spatio-temporal GWL dynamics from 2002 to 2024 by integrating GRACE/GRACE-FO-derived GWS, four static topographic-geomorphic factors, thirteen satellite-based hydro-environmental variables, and ground-based observations, with field verification. Among 39 model realizations using four algorithms, Random Forest consistently outperformed Support Vector Regression, Artificial Neural Network, and Extreme Gradient Boosting, achieving robust percentage-based performance metrics that are operationally acceptable (NRMSE = 15.01%, NMAE = 11.5%, NMSE = 0.02), and strong agreement with observed time series at most wells ($r > 0.6$). A two-step gap-filling strategy was applied to generate continuous pre- and post-monsoon GWL simulations, indicating that pre-monsoon conditions best represent long-term variability and depletion. The results identify critical depletion hotspots in Durgapur, the Kataberia coalbed methane fields, and north-eastern mining areas, with a maximum long-term decline rate of 13.4 cm yr⁻¹. Reconstructed depletion patterns exhibit a strong spatial correspondence with (Small Baseline Subset Interferometry) SBAS-InSAR-derived land subsidence and persistent behavior, as indicated by Hurst exponents of fractal analysis, greater than 0.67. The proposed framework provides reliable quantitative information to support GW assessment, urban planning, hazard mitigation, and sustainable management in intensively mined regions.

Index Terms—Groundwater, Machine learning, Random Forest, Fractal, Time-series, SAR interferometry (InSAR).

I. INTRODUCTION

Groundwater (GW) is a fundamental component of the global freshwater system, sustaining ecosystems, agriculture, and (Corresponding author: Paresh Nath Singha Roy).

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domestic water supply. In India, nearly 85% of rural drinking water and about 88% of extracted GW are used for agriculture [1]. Rapid urbanization, industrial expansion, and increasing climate variability have pushed GW systems into a state of growing stress, marked by declining water tables, deteriorating quality, and reduced recharge [2]. These impacts are most acute in coal-mining basins, where large-scale subsurface disturbance alters natural hydrogeological processes. Open-cast and underground mining breaches confining layers, enhance aquifer drainage, and disrupt GW flow paths [3]. Coalbed methane (CBM) extraction further intensifies stress by deliberately and long-term dewatering coal seams [4], [5]. These processes collectively reduce aquifer storage, accelerate land subsidence, and degrade water quality. The Raniganj Coalfield, India's oldest coal-mining region, has been operated continuously since 1774 and lies within the Damodar Valley basin of eastern India. GW abstraction has significantly increased since the start of CBM extraction in 2007 [6] and the rapid urban growth surrounding Durgapur and Asansol. A large portion of this growth has been uncontrolled.

Several geodetic approaches, such as the computation of the factor of safety from pillar stability analysis [7], visco-elastic model development [8], InSAR, and Differential Global Positioning System (DGPS) [9], have been used to describe widespread surface deformation in this coalfield.

Conventional physics-based GW models require accurate aquifer parameters, which are difficult to constrain in structurally complex, mining-disturbed coalfields. However, machine learning (ML) offers a viable alternative for such data-limited systems. Random Forest (RF), in particular, has demonstrated strong performance in large-scale GW reconstruction using GRACE-derived datasets [10], [11]. However, its application in India's mining-impacted GW systems remains limited.

This study integrates GRACE and GRACE-FO terrestrial water storage data (2002–2024), multisource satellite datasets (Dynamic Features), topographic-geomorphic variables (static features), and in situ GW level (GWL) observations to reconstruct spatiotemporal variations. RF models are developed for pre- and post-monsoon periods and systematically benchmarked against Extreme Gradient Boosting (XGB), Support Vector Regression (SVR), and Artificial Neural Networks (ANNs). The derived GWL trends were compared with InSAR-derived surface deformation, and fractal analysis was also performed to assess the long-term persistence of GWL variations.

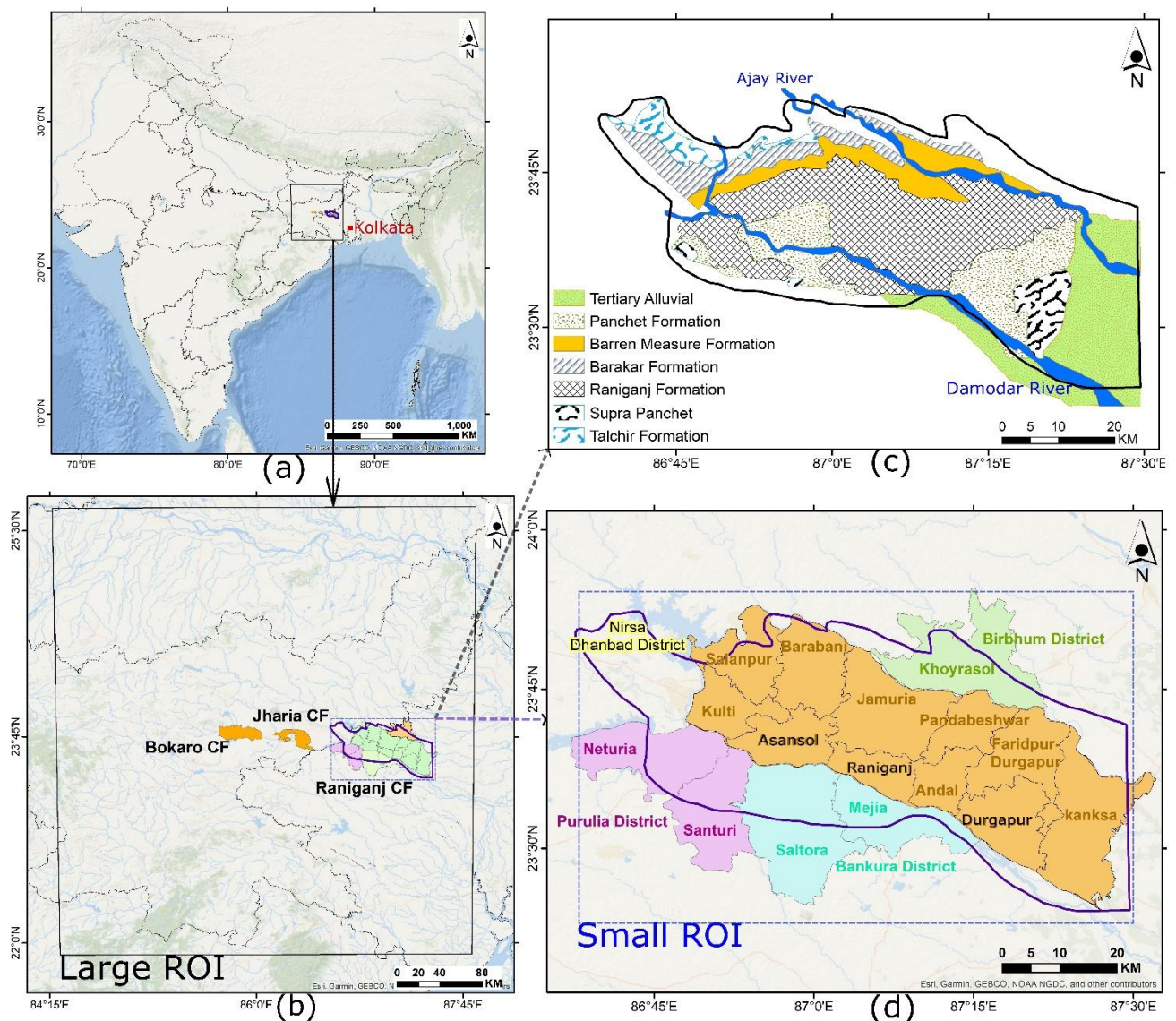


Figure 1 (a) Large region of interest (ROI) in relation to a map of India, which was used for the ML process. (b) Large ROI illustrating the spatial relationship between Raniganj, Jharia, and Bokaro coalfield. (c) Geological map of Raniganj coalfield. (d) Highlights the Small ROI that focuses on Raniganj, utilized for final analysis, along with outlining of the administrative block boundaries (modified after [9]).

This integrated framework provides a validated, long-term reconstruction of GWL for a highly stressed aquifer system. The study's framework is organized into distinct phases: Data Integration for multisource acquisition, Pre-processing for signal cleanup and outlier management, ML modeling for GWL reconstruction, and a final synthesis of InSAR and fractal analysis to validate subsurface trends.

A. Study Area

The Raniganj Coalfield (Fig.1) is situated in the Damodar Valley basin of eastern India, primarily within the Paschim Bardhaman district of West Bengal, approximately 182 km from Kolkata. The climate is monsoonal, with most rainfall occurring between June and September. The study uses a nested spatial framework, with the large **Region of Interest (ROI)**: approximately 153,779 km²) as the primary domain for ML model calibration. This allows for regional capture of the GRACE signal while keeping the model's outer boundaries far from the primary area of interest. This "interior-crop" approach eliminates edge effects and boundary leakage by cropping the

results to the Small ROI (approximately 5,665 km²) centered over the Raniganj coalfield. The ML training phase buffers the Small ROI with surrounding regional data.

Geologically, the coalfield is part of the Gondwana sedimentary basin and is composed of alternating sandstone, shale, and coal seams, resulting in a highly heterogeneous multi-aquifer system [12]. Basin development is controlled by Proterozoic tectonic structures and the formation of an intra-cratonic graben within the Damodar Valley [13], [14]. Aquifers primarily occur within fractured and weathered sandstones, while shales act as local confining units. Hydrogeologically, the region is intensively modified by open-cast and underground mining, CBM extraction, spoil dumping, and urban development. Continuous dewatering of coal seams, compaction of overburden materials, and the widespread development of impervious surfaces have substantially altered recharge and flow regimes. These combined controls render the Raniganj Coalfield a severely stressed and dynamically evolving GW system.

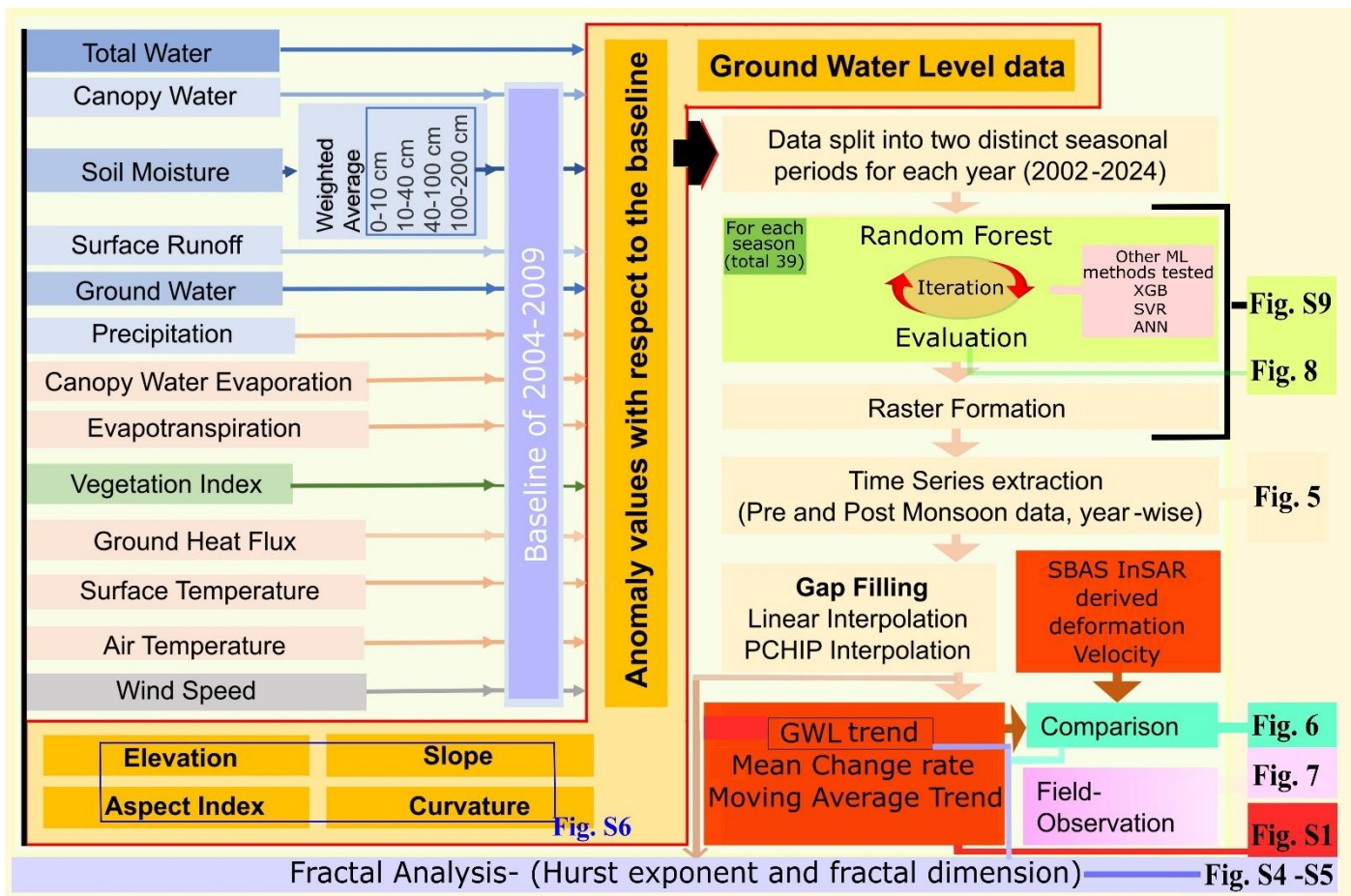


Figure 2 The workflow followed in this study. It showcases the transformation of dynamic variables into anomalies while retaining the static features derived from the DEM unchanged. Followed by data organization into two hydrological seasons: pre-monsoon (April-May) and post-monsoon (October-November) for each year, for each of the RF regression runs. This process resulted in the creation of predictive GWL maps; however, due to data gaps, 39 out of 45 maps were simulated. Finally, a time series was generated, and gap-filling techniques were utilized to address the missing data. The GWL trend based on the pre-monsoon season was then calculated, followed by comparison with InSAR-derived deformation velocity and Fractal analysis of the time series.

II. Data and Methodology

A. Dataset

A total of 17 input variables (Table S1) were used in the modeling process, which are of two types: 13 Dynamic Features, which include monthly data from 2002 to 2024 (listed as 1st to 13th), and 4 Static Features (listed as 14th to 17th). The Central Ground Water Board (CGWB)-derived GWL observations at existing monitoring wells were used as the target (dependent) variable for supervised model training, while the 17 environmental and hydrogeological parameters served as predictors.

1) Dynamic Features

The GRACE Terrestrial Water Storage (TWS) data offer a satellite-derived estimate of the total water stored in the land column, presented in a 0.25-degree grid. However, the mascons (Mass Concentration blocks) are estimated using a 1-degree equal-area grid, and the native resolution of the GRACE/GRACE-FO data is approximately 300 km [15]. To interpret the integrated TWS signal and its spatiotemporal variability, it is essential to examine the interrelationships among its key surface and subsurface hydrological components. Canopy Water (CW) reflects water stored in vegetation and influences interception losses, while Soil Moisture (SM) indicates soil water content, which is crucial for infiltration and plant uptake. Surface Runoff (Qs) represents overland flow and inversely correlates with recharge potential. The GWS parameter here is derived from GRACE/GRACE-FO, as will be

discussed in subsection 2.3.1. Evapotranspiration (ET) represents the combined loss of water through evaporation and transpiration, thereby reducing the water available for recharge. In contrast, surface temperature (ST) and air temperature (AT) regulate evaporation rates and influence soil moisture dynamics. Wind speed (WS) enhances evapotranspiration by increasing atmospheric demand, and ground heat flux affects subsurface temperature and biological activity. Canopy evaporation (CE) quantifies direct water loss from vegetation, reducing the volume available for infiltration and recharge [16]. Rainfall (PT) is the primary source of recharge. Its data are from CHIRPS-2.0 at 0.05° resolution [17], while the MODIS-derived Normalized Difference Vegetation Index (NDVI) serves as a proxy for vegetation health and transpiration demand [18].

2) Static Features Dataset

FABDEM is a high-resolution global elevation model (30 m) that removes surface features such as vegetation and structures to accurately represent bare-earth topography [19]. Thereby, using this DEM, four terrain attributes: elevation, slope, curvature, and aspect index (Fig. S6), were derived. Elevation acts as the hydraulic head that controls GW flow, recharge, and discharge patterns. Slope affects elevation change, impacting runoff, infiltration, and erosion. Steeper slopes increase runoff, while gentler slopes enhance recharge. Curvature influences the shape of the terrain and, in turn, localized recharge. Curvature influences the shape of the terrain and, in turn, localized recharge.

Aspect index (Normalized aspect into 0-1 scale) affects slope orientation, which influences microclimate, evapotranspiration, and soil moisture distribution, leading to variations in recharge potential [20], [21], [22].

3) Data availability

All the data utilized in this study are publicly available. For land-surface data, the GLDAS dataset is available from NASA's Goddard Space Flight Center (<https://ldas.gsfc.nasa.gov/gldas>). The CHIRPS data can be accessed at the University of California, Santa Barbara's website, which provides detailed precipitation data (<https://www.chc.ucsb.edu/data/chirps>). Additionally, data from the GRACE satellite mission, which monitors water resources, can be found on the University of Texas at Austin's website (<http://www2.csr.utexas.edu/grace>). For Synthetic Aperture Radar (SAR) data, users can visit the Alaska Satellite Facility's search portal (<https://search.asf.alaska.edu/>). Lastly, well data from the CGWB of India is available through the India Water Resources Information System website (<https://indiawris.gov.in/wris>).

B. Pre-processing

To ensure methodological rigor and comparability across datasets, a systematic pre-processing protocol was established before developing the model. All satellite-derived raster datasets used in the machine learning process were resampled and re-gridded to a uniform 0.01-degree grid using bilinear interpolation. Fig. 2 illustrates the subsequent steps, which involve aligning all dynamic variables with the GRACE TWS Anomaly's baseline (the mean of 2004—2009) [23].

For each dynamic variable, the baseline mean was computed over this period, and subsequent values were transformed into anomalies by subtracting the baseline from each monthly observation, ensuring that all variables were standardized against a consistent hydrological reference. To derive GWS-data from GRACE/GRACE-FO-derived total water storage (TWS), the following formula was used:

$$GWS = TWS - (CW + SM + Qs) \text{ -----Eq.1}$$

In this equation, SM is the weighted average of four layers (0–10 cm, 10–40 cm, 40–100 cm, and 100–200 cm). Qs was converted to monthly data by changing it from a 3-hourly format. To align with TWS's linear dimension, CW, SM, and Qs were converted to cm from kg/m² by multiplying by 0.1 [24], [25].

Following the data were temporally sorted into two distinct hydrological seasons to capture the region's recharge and discharge dynamics, pre- and post-monsoon. The pre-monsoon season was defined using data from April and May, while the post-monsoon season utilized data from October and November. When data from both months within a season were available, their average was used to represent the seasonal value; if only one month's data was available, that value was used solely for the season. This seasonal aggregation minimized transitional noise and aligned the dataset with key hydrological processes. Therefore, a total of 39 distinct Integrated datasets were generated out of 45, covering the period from pre-monsoon 2002 to pre-monsoon 2024. The missing data points are attributed to gaps in GRACE/GRACE-FO data or the unavailability of well data, including those caused by the pandemic in 2020 and the 2021 pre-monsoon season.

C. ML modeling

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The strength of the RF method lies in its ability to avoid overfitting, which is particularly valuable in areas with limited data, like the study area, Raniganj. Here, GW dynamics are influenced by a variety of factors, including mining materials, mining methods, and anthropogenic loading, and by a lack of dense, high-resolution ground truth data. RF works by creating an ensemble of decision trees from different random samples of the data. Each tree splits the feature space to minimize differences in the target variable within the subsets it creates. For regression tasks, this means that each leaf node of a tree shows the average target value for the samples it includes. To make predictions, RF averages the results from all the trees. This collective approach, often referred to as the "wisdom of the crowd," helps improve accuracy and prevents the model from becoming overly tailored to any specific dataset [26], [27].

Consequently, to implement the RF, as a first step, all input features were standardized using z-score normalization to enhance comparability and model stability. The dataset was split into 80% for training and 20% for testing [28]. To minimize overfitting and provide reliable performance estimates, a five-fold cross-validation scheme was embedded within a recursive, multi-stage optimization loop. To prevent optimization bias and structural overfitting within this nested framework, the key hyperparameters for the RF model were systematically tuned using a focused, regularized grid search. The parameter envelope was explicitly bounded to discrete, stable values: the number of decision trees was set to either 200 or 300, the minimum leaf size was restricted to 5 or 10, and the number of split predictors was evaluated at either the square root of total features or half of the feature pool.

Rather than executing an unconstrained automated sweep—which risks memorizing localized data splits—this controlled parameter envelope functions as a vital structural regularizer. It ensures that the model configurations remain stable and physically consistent across the highly repetitive outer wrapping iterations without over-optimizing to high-frequency data noise.

This approach produced a well-calibrated, robust Random Forest model that maximized generalizability and minimized overfitting. This phenomenon of enhancing generalizability while reducing overfitting can be directly observed in Fig. S9, which shows the actual versus predicted values for each data point and the final predicted GWL for each season over the time period.

The predicted GWL can be expressed as,

$$GWL_{pred} = \frac{1}{n} \sum T_i(x) \text{ -----Eq.2}$$

where n is the total number of trees and T_i(x) is the prediction from the ith tree. This study compared four machine learning models, RF, XGB, SVR, and ANN, for predicting GWLs. Among these, the RF consistently achieved the best evaluation metrics across all years analyzed, outperforming XGB, SVR, and ANN. Using model-predicted GWL data from 39 occurrences, we generated time-series data (Fig. 5) and observed that 6 points contained missing values. To create the temporal profiles, model-predicted groundwater levels were obtained from the 39 independent seasonal spatial models and assembled into localized (Fig. 5). Six specific observation nodes displayed gaps due to deficiencies in GRACE/GRACE-FO data or the unavailability of

well data, including those resulting from the pandemic in 2020 and the pre-monsoon season of 2021. Each of the 39 models functions as an entirely isolated, discrete temporal snapshot; hence, the train-test splitting was conducted within separate spatial domains rather than across a continuous temporal sequence. This separate-snapshot design automatically mitigates temporal data leakage—where a model is trained at time T and tested at time $T+1$ —as sequential temporal steps are never concurrently exposed to any individual model instance.

To rectify these data deficiencies, we used a two-stage imputation methodology: first, we applied linear interpolation to address momentary data omissions. Subsequently, we used polynomial harmonic interpolation (PCHIP) [29] smoothing to reinstate the seasonal patterns and trends. This hybrid methodology aligns with established methods in seasonal GW modeling, where preserving data integrity during absent months is crucial for accurate forecasts. By maintaining spectral properties, PCHIP smoothing alleviates the oversimplification that may arise from linear interpolation. Afterward, three kinds of GWL change-based maps were constructed with respect to the pre-monsoon GWL of all the points (1 km spacing) that fall in the small ROI, which are of cm yr^{-1} unit, are trends, Mean Annual Changes (Year-over-year), and Moving Average trends (3-year interval) (Fig. S1).

D. InSAR and fractal analysis

Selecting the Small Baseline Subset (**SBAS**) approach for monitoring active mining environments is a strategic choice driven by the need to maintain signal coherence in a landscape that is physically changing every day. While **Persistent Scatterer InSAR (PS-InSAR)** is the gold standard for urban stability, its reliance on "point-like" stable reflectors (such as buildings or rock outcrops) often results in total data "blackouts" in excavated areas where the surface is constantly being reworked. The SBAS method creates a time series of interferograms using small temporal and spatial baselines to reduce decorrelation and atmospheric noise. Key steps include coregistration of SAR images, generating interferometric pairs, phase unwrapping, and time-series inversion for vertical ground displacement analysis [30]. The InSAR dataset used in this study consists of 64 descending Sentinel-1 images acquired between 2017 and 2023. This dataset was earlier reported by [9] for a limited spatial domain. In the present study, the same dataset is reprocessed for an expanded study area using the SBAS technique, yielding 411 interferograms with spatial and temporal baseline thresholds of 100 m and 250 days, respectively. To align with water management (USGS) frameworks that treat depletion as a positive loss, we adopt a 'downward-positive (+), upward-negative (-)' sign convention. Accordingly, InSAR-derived velocities and model-predicted GWL data are signed so that increasing positive values directly represent greater ground subsidence and water-table drawdown.

Fractal analysis of the model-generated GWL time series was performed using the Hurst–Mandelbrot method. The analysis begins by segmenting a time series of length N into n contiguous, non-overlapping sub-periods of size k , where the initial window size is typically set to $k=5$ observations to ensure sufficient data for reliable statistical estimation. For each block

at a given scale k , the algorithm computes the mean

$$\mu = \frac{1}{k} \sum_{i=1}^k x_i \quad \text{-----Eq.3}$$

and constructs the mean-adjusted cumulative deviation series

$$y_t = \sum_{i=1}^t (x_i - \mu) \text{ for } t = 1, 2, \dots, k \quad \text{-----Eq.4}$$

from which the range is obtained as $R(k) = \max(y_1, \dots, y_k) - \min(y_1, \dots, y_k)$ -----Eq.5

which is then "rescaled" by the local standard deviation

$$S(k) = \sqrt{\frac{1}{k} \sum_{i=1}^k (x_i - \mu)^2} \quad \text{-----Eq.6}$$

to give the rescaled range to quantify long-term dependence

$$(R/S)_k = \frac{R(k)}{S(k)} \quad \text{-----Eq.7}$$

For each outlet, missing values were removed, and the series was divided into progressively larger non-overlapping blocks. Within each block, the cumulative deviation from the mean was calculated, normalized by the standard deviation, and averaged to obtain R/S values for each scale. Plotting $\log(R/S)$ against $\log$ (block size) provided the Hurst exponent (H) as the slope of the linear scaling region, from which the fractal dimension (D) was computed ($D = 2 - H$).

III. Results

As mentioned in subsection A. *Study Area*, the small ROI is an interior crop of large ROI, here, only the results and evaluation metrics related to the 2002 pre-monsoon have been showcased (Fig. 3). For all the years and corresponding metrics along with the residuals vs. predicted plots, please refer to Fig. S9, where results for each of the seasons are presented in the same order as in Fig. 3.

Before running a regression model, it is crucial to examine the correlation matrix, which highlights feature dependencies by showing how strongly two variables are connected.

A large positive correlation suggests redundancy, but a negative correlation is equally vital, indicating an inverse relationship where one variable rises as the other falls. Hydro-meteorological and geomechanical variables show that the region is dominated by dynamic, non-stationary physical processes. Using a single global correlation model might mask significant changes in land-surface feedback loops between pre- and post-monsoon cycles. Bivariate correlation studies reveal temporal instability in feature connections, diagnosing these shifting regimes (see the correlation matrices in Fig. S9). Since GWS is derived directly from the TWS mass-balance equation, a strong positive correlation is always observed. The 2002 pre-monsoon period was moisture-limited, with negative relationships between Canopy Water (CW) and Surface Temperature (ST) and between ST and Evapotranspiration. The substantial negative relationship between Wind Speed (WS) and Soil Moisture indicates that aerodynamic forces became the primary cause of moisture loss by the 2002 post-monsoon. These physically intricate relationships emerge over 22 years, reflecting the landscape's response to excavation. GW was adversely connected with Wind Speed in the 2023 pre-monsoon, whereas ET was strongly favorably correlated with Canopy Evaporation [31], [32]. By the 2024 pre-monsoon, typical patterns broke down, with strong GW-WS negative correlations and a jump in ST-ET negative coupling. These oscillations show that the "weight" or physical importance of any environmental parameter fluctuates over time [33]. A single model for the dataset would "average out" these seasonal fingerprints, reducing forecast accuracy and missing high-magnitude subsidence occurrences during transitional

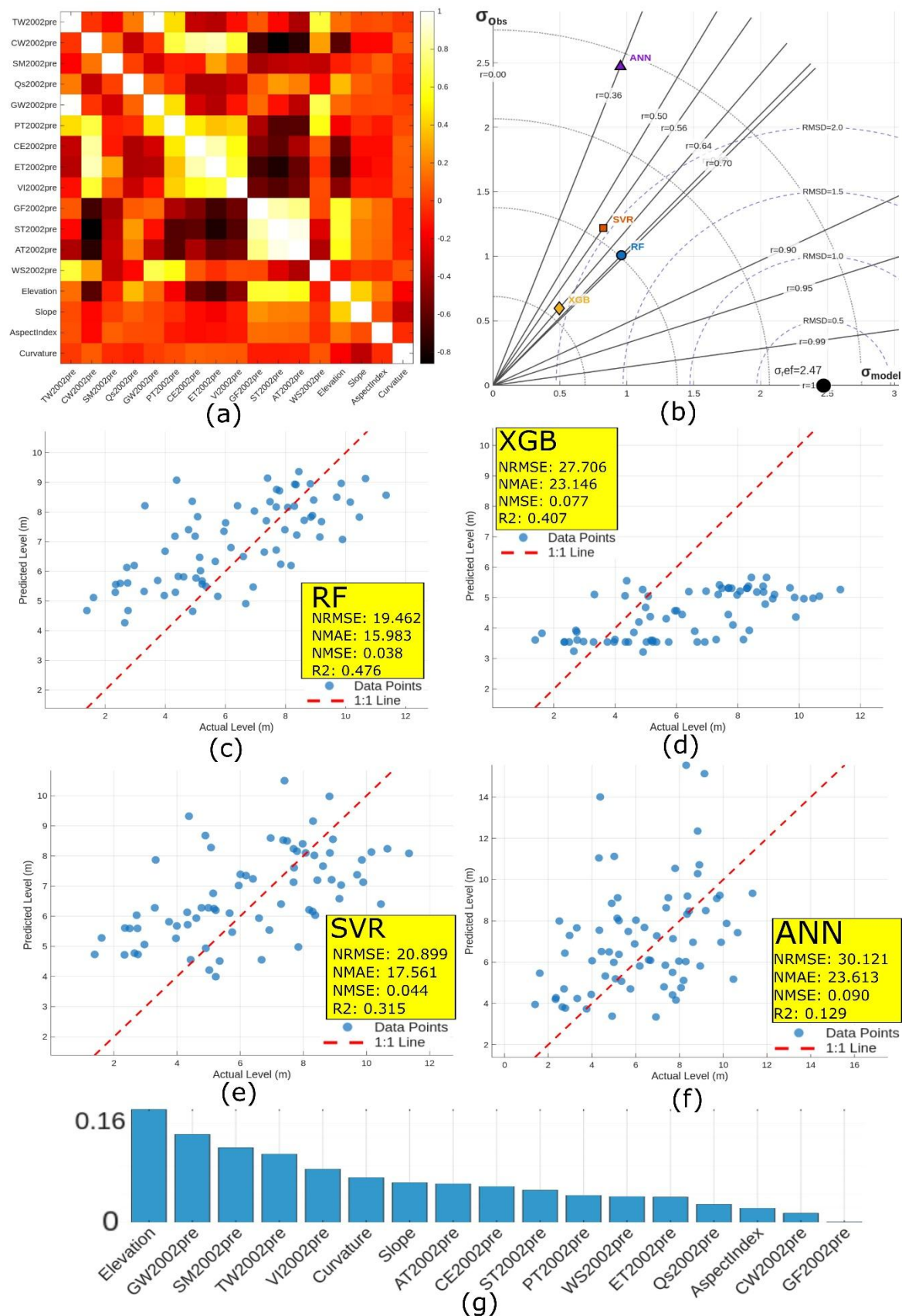


Figure 3 (a-f) Results, comparison, and evaluation of the ML models (2002 pre-monsoon) executed: (a) The correlation matrix among the various hydrogeological and environmental parameters utilized in the machine learning model for GWL estimation. The matrix highlights the degree of linear association between each pair of parameters, aiding in the identification of key influencing factors and potential multicollinearity within the dataset. (b) Taylor diagram for the ML model results. (c-f) Actual versus predicted graphs of the test results for the models (c-f - RF, XGB, SVR, and ANN), along with the values from the evaluation factors calculated for each individual model, are presented as an overlay image in the bottom right corner. (g) Feature importance plot from the RF (the dynamic parameters' names have two letters for the dynamic parameter's name itself, followed by YYYY format for the corresponding year, like here it's 2002, followed by pre, depicting pre-monsoon or post, depicting post-monsoon here and in the corresponding rest of the figures in Fig. S9).

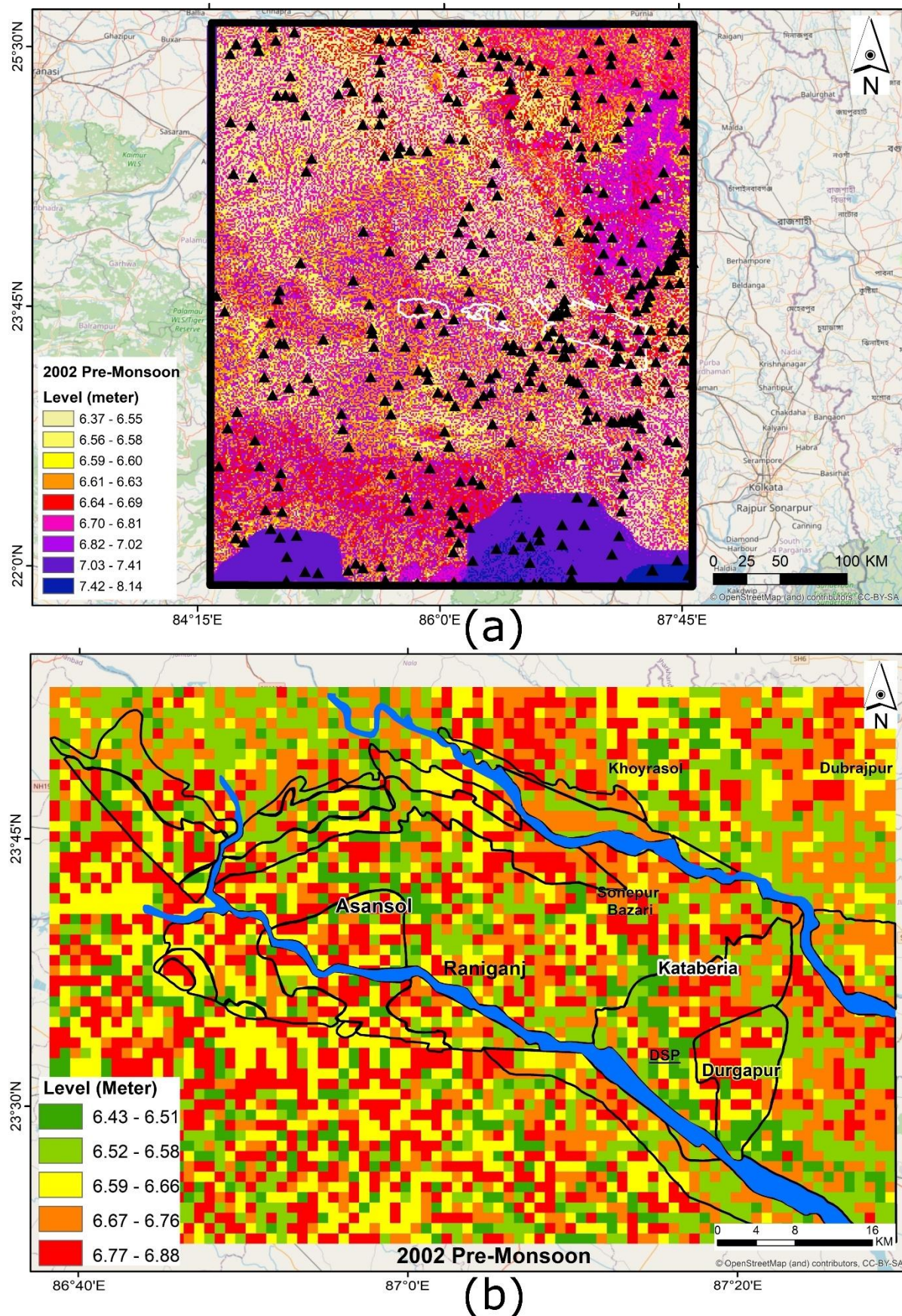


Figure 4 (a) Spatial distribution of estimated GWL (large ROI) of 2002 pre-monsoon, with the outlines of Raniganj, Jharia and Bokaro coalfield, along with the GW stations which were used in final training, are shown in black triangles. Hereby, the classification is geometrical with 9 classes. (b) Spatial Distribution of Estimated GWL (Small ROI) of 2002 pre-monsoon, with the outlines of Raniganj coalfield's geological (Formation) boundary and rivers. Hereby, the classification is geometrical with 5 classes.

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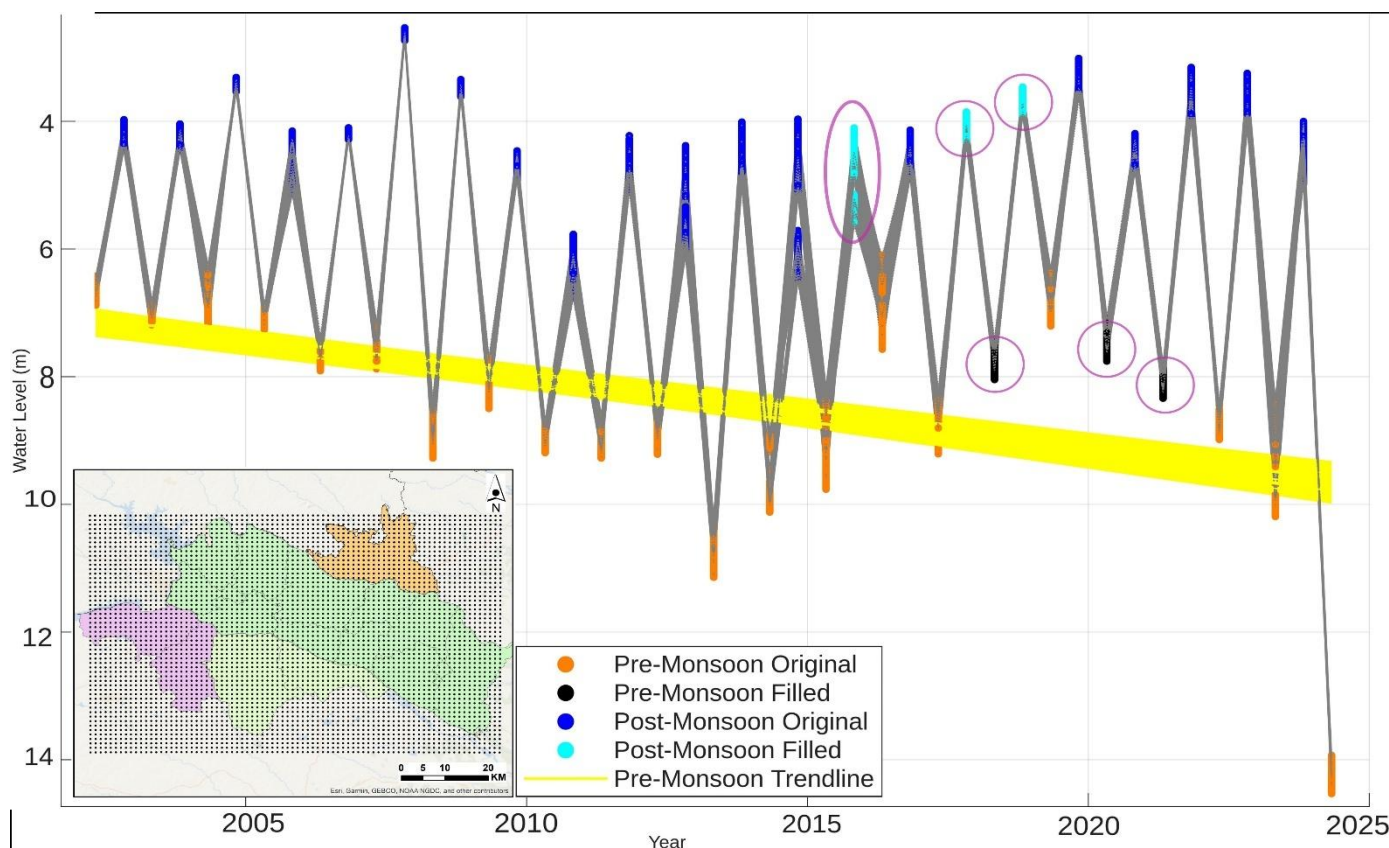


Figure 5 Constructed time series from all the virtual monitoring points (54 *82= 4428) spaced 1 km apart in the small ROI, as showcased in the overlay at the bottom left, with the district boundaries associated with Raniganj Coalfield. The filled gaps are indicated in the legend under the "Filled" label, which is also marked with red circular lines in the accompanying picture. The figure also illustrates the trends (yellow lines too close to be seen) based on the pre-monsoon period, providing a dynamic perspective on GWL for each point.

climatic phases [34], [35], [36].

The model's spatial partitioning was isolated across independent six-month seasonal snapshots over 22 years to avoid spatiotemporal autocorrelation in localized machine learning frameworks, enabling the learning of dynamic physical drivers rather than site-specific temporal signatures. Spatial block cross-validation is theoretically rigorous for large, data-dense regions but is unsuitable for highly localized and sensitive areas such as the Raniganj Coalfield.

The GRACE satellite footprint scale shows that the Raniganj Coalfield covers less than 2.5% of the regional area. An intrinsically sparse and physically limited groundwater observation well network monitors this vital mining zone. Broad geographic blocking (such as Spatial Block Cross-Validation) would isolate and remove critical, unrepeatable localized hydrogeological characteristics from the model's training pool in such a data-scarce, highly diverse mining environment. This would blind the machine learning framework to the coalfield's specific structural and anthropogenic causes, reducing the model's localized representative ability and predictive validity.

Our iterative randomization methodology and tight separation of the recursive outlier-removal phase from the initial training split minimize information leaking and overestimation [37].

The pipeline repeats up to 20 outer iterations of candidate sample removal across several removal fractions (5%, 10%, and 15%) throughout 500 train-test resamplings to reliably investigate diverse model configurations and data-cleaning stages. The model's spatial stability is confirmed by this extensive outer wrapping structure, and the iteration with the highest generalized accuracy was used [38].-

To compare how well the machine learning models (RF, XGB, SVR, and ANN) perform against actual observed data, we created a Taylor diagram [39] (Fig. 3b). This diagram serves as a useful visual tool to illustrate the differences and strengths of each model in a clear and concise way, comparing the correlation coefficient (r), root mean square deviation (RMSD), and standard deviation (σ) of the simulated and observed fields, making model evaluation more intuitive. The radial axis represents the standard deviation of model predictions, and the angle from the x-axis indicates the correlation with observed data. The distance from the reference point (which represents the observed data) reflects the centered RMSD, indicating how closely the model matches the observed patterns and variability. This makes Taylor diagrams especially useful for comparing multiple models in regression tasks.

We evaluated the model output using standard statistical metrics, including Normalized Root Mean Squared Error (RMSE) or NRMSE, Normalized Mean Absolute Error (MAE) or NMAE, Normalized Mean Squared Error (MSE) or NMSE, and the R^2 . The normalized formulas used for these metrics are provided below for clarity.

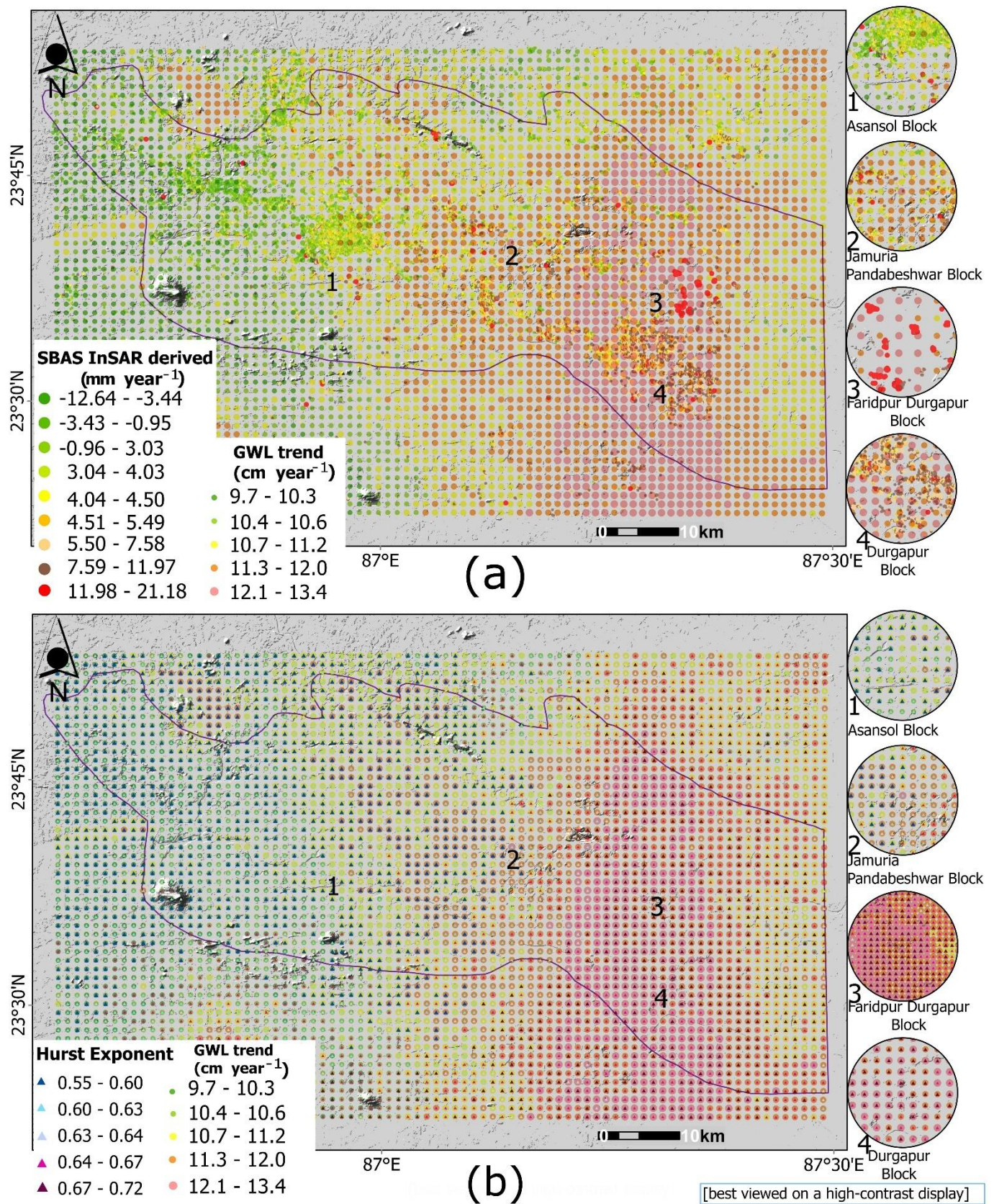


Figure 6 (a) SAR-derived deformation velocity superimposed upon the GWL trend Map. Sign Convention: Positive (+) for downward, negative (-) for upward. (b) The figure below showcases the Hurst exponent.

$$\text{NRMSE} = \frac{\text{RMSE}}{(\frac{y_{\max} - y_{\min}}{\text{MAE}})} \quad \text{-----Eq.8}$$

$$\text{NMAE} = \frac{\text{MAE}}{(\frac{y_{\max} - y_{\min}}{\text{MSE}})} \quad \text{-----Eq.9}$$

$$\text{NMSE} = \frac{\text{MSE}}{(\frac{y_{\max} - y_{\min}}{2})^2} \quad \text{-----Eq.10}$$

It is noteworthy that the evaluation metrics for all outputs remained within accepted thresholds for modeling performance, indicating reliable predictions. Fig. 3(c-f) shows the actual vs. predicted plots and presents evaluation metrics for the 2002 pre-

monsoon for each model.

The dynamic shifts in feature importance (Fig. 3g) —where elevation (0.16) and soil moisture dominate the 2002 pre-monsoon, whereas GRACE-derived TWS (~0.12) and surface temperature take precedence in the post-monsoon of the same year (Fig. S9) —confirm that the model successfully captures seasonal physical transitions. By allowing the importance hierarchy to reconfigure from topographic and moisture drivers to regional storage and thermal forcing, the RF framework maintains high predictive fidelity that a static, non-segmented global model would otherwise average out and obscure.

Fig. 4 shows the predicted GWL for the pre-monsoon period in 2002 for the large and small ROIs. The time series was generated for all virtual monitoring points in the small ROI (Fig. 5). Thereafter, the three pre-monsoon-based parameters were generated: the first, GWL trend, represents the slope of a first-degree polynomial (linear regression) fitted through the time series of pre-monsoon water levels using year as the independent variable. The second metric, Mean Annual Change rate, captures the average year-to-year change in pre-monsoon levels by calculating the difference between consecutive years and averaging them (Fig. S2).

The third metric, Smoothed Moving Average trend, applies a moving average filter (3-year window) to the annual change series, providing a less noisy, smoothed rate of GW movement, computed as the mean of these smoothed values (these 3 pre-monsoon-based simulated GWL derived parameters with respect to the study area are shown in Fig. S1). Afterwards, the derived trends are superimposed with SBAS-InSAR-derived surface deformation velocities (Fig. 6a), enabling a comparative analysis of subsurface hydrological dynamics and surface movement patterns. Three vertical histogram panels (Fig. S3) illustrate the spatial distribution of long-term GWL-derived parameters across all virtual monitoring points spaced 1 km apart in the small ROI. The time series were subjected to fractal analysis. The spatial distribution of the Hurst exponent, along with the trend, is shown in Fig. 6b, while the corresponding fractal dimension map is presented in Fig. S4. Histograms of both H and D related to fractal analysis are illustrated in Fig. S5.

IV. Discussion

A. Hydrogeological Controls and Field Evidence

The Raniganj Coalfield receives the majority of its annual rainfall from the Southwest Monsoon, spanning June to September. However, effective rainfall-to-recharge conversion is often limited due to complex hydrogeological and anthropogenic conditions. Early monsoonal rainfall is largely absorbed by dry, desiccated surface soils, while subsequent rains tend to generate runoff due to the compacted surface and low-permeability substrates. The dominance of lateritic soils in the region, characterized by poor infiltration capacity, further restricts vertical percolation of precipitation [40]. Field observations (Fig. 7) conducted in northern Pandabeshwar, Faridpur–Durgapur, and southern Khoyrasol identified significant instances where spoil dumps, overburden material, and disturbed pit floors act as barriers to GW infiltration. These areas are dominated by open-cast and underground mining operations that alter surface hydrology and compact the

subsurface, diminishing the formation of natural recharge channels. The problem is further magnified by sprawling urban settlements and impervious surfaces—particularly in Durgapur—which significantly reduce infiltration and promote surface runoff.

B. GW Depletion, Spatial Concordance with Deformation and Error Metrics

Additional detailed photographic documentation from these sites is available in a previous field study reported in Fig. 10 of Ghosh et al., 2024 [9]. Field observations are strongly supported by modeling results. Spatial analysis of GWL change trend revealed vulnerable zones of high depletion, with calculated rates ranging from 12.1 to 13.4 cm yr⁻¹. These areas coincide with intense anthropogenic activities. In the area near Kataberia, industrial-scale CBM well dewatering is likely exacerbating the drying of shallow aquifers. Similarly, extensive open-cast mining in the northeast sector (Pandaveswar–Sonepur Bazari) contributes to aquifer drawdown, while the Durgapur Urban Region faces compounded stress from municipal extraction, overlapped with impermeable surfaces that limit recharge. These hydrological changes are not occurring in isolation. A spatial overlay of Random Forest–derived GWL change trend estimates with SBAS InSAR data revealed strong correspondence between subsidence and zones of significant GW withdrawal. In areas of maximum overlap, the InSAR-derived deformation velocity reached up to approximately +21 mm yr⁻¹. This strong hydro-mechanical concordance between differential land motion and GW velocity signals a coupled subsurface response, consistent with observations in other terrains.

The Random Forest (RF) model was chosen over the Extreme Gradient Boosting (XGB) model after a thorough evaluation of their physical predictive capabilities, especially in this mining environment as it demonstrated a more robust and reasonable approximation (Fig. S9). Although R² values may indicate statistical parity, such as in the 2023 pre-monsoon data when RF attained 0.668, and XGB followed closely at 0.628, this measure alone is insufficient for evaluating geomechanical model physical dependability. Measurements that penalize distinct model failures are physically significant. Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) quantify millimeter displacement deviations. In dynamic terrains, the RMSE is particularly telling. XGB's significantly higher RMSE values (6.693 in 2024 pre-monsoon compared to 1.775 for RF) indicate that the boosting algorithm is prone to extreme "outlier" errors, which could lead to dangerous underestimations of slope stability or false alarms. Efficiency metrics such as the Nash-Sutcliffe Efficiency (NSE) and Kling-Gupta Efficiency (KGE) demonstrate the RF model's superiority. These measures assess how well the model captures the mean and variability of observed displacement to provide a deeper physical explanation. During the 39 observation locations, XGB had negative NSE values, including -0.264 in the 2002 pre-monsoon and -2.312 in 2024. A negative NSE indicates that the model's predictions are less accurate than using the mean of the observed data as a constant predictor, making the XGB model unsuitable for localized surveillance at certain times. Even after the excavation reduced the InSAR signal, the RF model retained positive NSE and KGE values. KGE results show that RF preserves the "spread" or variation of displacement, while XGB over-smooths data. The normalized measures (NRMSE and



Figure 7 Field Observation: (a, b) In the water-filled opencast mine, the pit floor is compacted and has low permeability, which hinders GW recharge. (c) An opencast mine with visible semi-filled waste materials and water. (d) The excavation of elevated coal seams has significantly altered the landscape. (e) Large mine spoil dumps are forming new artificial landforms in the area.

NMAE) enable an objective comparison of error scales over seasonal cycles, proving the durability of the Random Forest approach. The 2024 pre-monsoon NRMSE for XGB reached 67.949%, while RF stayed at 18.023%. This shows that the RF model generalizes better at capturing complex, nonlinear interactions between excavation factors and surface response, without overfitting like gradient boosting in noisy settings. Although XGB showed excellent correlation, such as the 2023 post-monsoon R^2 of 0.845, the related KGE was just 0.365, much lower than the RF KGE of 0.472. Even while tracking the broad trend, XGB fails to physically depict ground movement. Thus, the RF model is the best instrument for this investigation, as it is stable, physically realistic, and robust to errors for monitoring continually damaged mining landscapes.

We provide the mean metrics of RMSE, NRMSE, MAE, NMAE, MSE, NMSE, R^2 , Adjusted R^2 (Adj R^2), Nash-Sutcliffe Efficiency (NSE), and Kling-Gupta Efficiency (KGE) from 39 RF models in Fig. 6, where each of the metrics corresponding to each of the RF models is shown. The RMSE value of 2.29 m indicates appropriate forecast errors for normal GW changes. Similarly, the NRMSE of 15.01% is operationally acceptable, falling within the standard hydrological modeling threshold of

<20%. The MAE of 1.73 m and NMAE of 11.5% confirm the model's consistency. The MSE of 5.46 m^2 suggests little variance from observed values, while the NMSE of 0.02 indicates great accuracy. The values of R^2 (0.48) and adjusted R^2 (0.47) are close, indicating that the independent variables in the model are meaningful and contribute meaningfully to explaining the variance in the dependent variable. In data-limited or complex aquifer systems, efficiency measures such as NSE (0.39) and KGE (0.35) demonstrate good agreement between simulated and actual results. Although each of the model-generated raster had acceptable metrics, we still sought to examine the correlation (r) between the generated time series and the available station data. This analysis is illustrated in Fig. S7-S8, showing satisfactory results with the majority of wells having values greater than 0.6. While well near the major depletion zones (GWL trend >12.1 $cm\ yr^{-1}$) like Batikar, Chunpara, Khagra Pz, Khayrasol, Tarasingh Bridg, Dakshin Khanda has good correlation, some of the surrounding wells with mostly lesser data have lower correlation like Mejia 2, Dshbandhunagar, Kanksa, Dobarna (New Kenda).

[11] tackled the coarse ~100–300 km spatial scale of GRACE TWS data by deploying an RF-based downscaling approach to achieve 1 km resolution in Morocco, enabling far more detailed tracking of drought dynamics. Their validation strongly

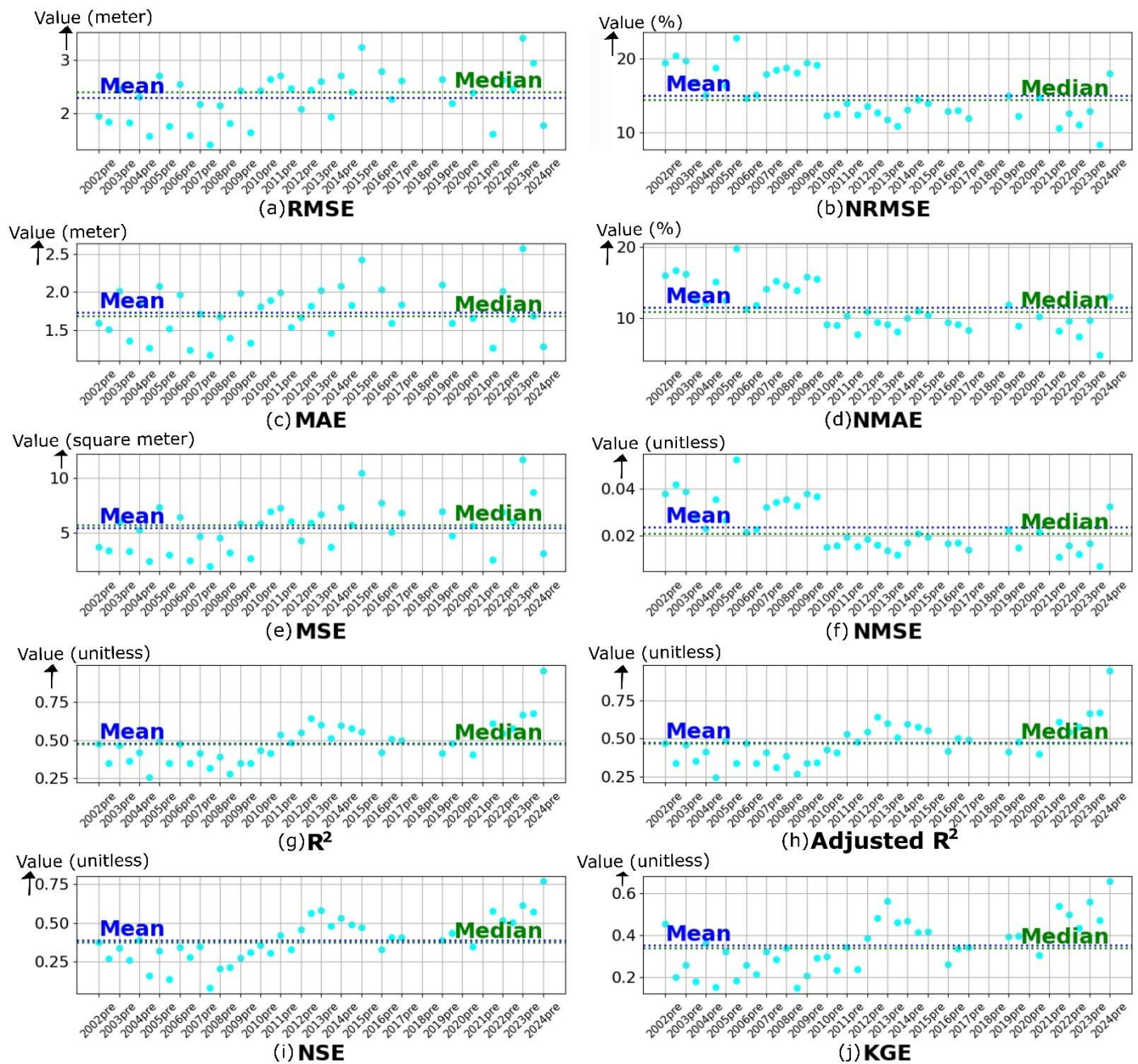


Figure 8 (a-j): Error metrics of the RF models.

emphasized not just in-situ and statistical agreement, but also the relevance of NRMSE, NMAE, NMSE, and NSE, with low RMSE (0.82 cm) and MAE (0.57 cm) highlighting the method's predictive accuracy. Similarly, [10] used RF to downscale GRACE-derived GW storage to 5 km in California's Central Valley, illustrating that NSE provides more robust insight into nonlinear system skill than R^2 . Koch et al. combined RF with physical GW models at 50 m resolution in Denmark, focusing on GW flood risk and evaluating using MAE (76 cm) and R^2 (0.56), while also highlighting model interpretability and uncertainty [41]. As these studies show, while R^2 remains a familiar metric, its limitations—especially insensitivity to additive/proportional bias or low-variance bias—mean that comprehensive model evaluation in hydrology should incorporate NRMSE, NMAE, NMSE, NSE, and KGE as essential, scale-independent indicators [42], [43], [44].

C. Physical Consistency and Model Validity

By leveraging surface deformation and environmental variables as proxies for subsurface changes, the Random Forest model effectively overcomes regional data gaps. While frameworks like Physics-Guided Machine Learning (PGML) [45], [46] represent an important path for embedding explicit conservation laws, our multi-feature configuration is firmly grounded in foundational water-energy dynamics.

The model's changing feature importances illustrate its ability to reflect dynamic physical processes across distinct seasonal phases, rather than merely correlations. The framework operates separate models at six-month intervals, with each execution serving as a diagnostic snapshot of evolving basin hydraulics. During the dry pre-monsoon period, topographically influenced

factors such as elevation predominate in the model, accompanied by relatively stable and less influential soil moisture profiles, effectively delineating a landscape regulated by gravity-driven lateral drainage and baseflow recession. In contrast, the model dynamically adjusts to the post-monsoon snapshot, where GRACE-derived parameters, precipitation, and evapotranspiration data predominantly influence the feature weights, accurately monitoring the extensive regional mass-balance alterations and hydro-climatic fluxes induced by significant monsoon recharge.

Auxiliary variables further validate this snapshot architecture as physical proxies rather than arbitrary statistical regularizers. Ground heat flux dictates shallow vadose zone desiccation and evapotranspiration, controlling the soil moisture deficit that limits subsequent rainfall penetration. Concurrently, the aspect index captures topographically driven solar radiation anomalies, governing localized runoff retention versus diffuse infiltration through the Gondwana basin's exposed fracture networks. By anchoring shifting mathematical weights to these established ecohydrological mechanisms across isolated time windows, the framework proves it interprets genuine hydrogeological responses rather than non-physical noise.

D. Limitations

D.1. Spatial Downscaling Fidelity and Local Hydrogeology Warning

The incorporation of coarse-scale TWS and GWS data (0.25°) is methodologically complemented by the integration of high-resolution static and dynamic characteristics, including 30 m FABDEM elevation data and 1 km MODIS NDVI. This design utilizes the regional GRACE signal to utilize regional gravitational trends to guide local water storage approximations. However, the ML framework cannot natively simulate true micro-scale local hydrogeological features from a coarse regional gravitational grid.

At a 1 km grid scale, intense localized mining, cones of depressions, small fault structures, or individual pumping station footprints will deviate drastically from the broader regional signals captured by GRACE. Therefore, readers are explicitly cautioned against treating these downscaled data pixels as precise field observations or "ground truth" at the borehole level. These 1 km maps serve solely as spatial proxies of regional trends rather than independent, field-validated hydrogeological measurements.

D.2. Data Constraints and Monitoring Gaps

The Damodar, Ajay, Kunur, and Kopai rivers drive the region's complex hydrological network, which was initially simulated using physically based models such as SWAT and MODFLOW. Due to data constraints, such as sporadic river gauge observations, absence of continuous river flow data, and lack of high-resolution soil characteristics, these frameworks could not be robustly applied. Complex anthropogenic factors, such as Damodar Valley Corporation (DVC) regulatory water releases and confidential pumping rates or abstraction well locations for local coal and CBM operations, are difficult to parameterize without restricted internal operating data.

A data-driven methodology reduced these data shortages, although spatiotemporal limits remain. GRACE/GRACE-FO mission gaps and pandemic-related field disturbances during the 2020 and 2021 pre-monsoon seasons caused missing data

points in the 39 seasonal datasets.

E. Future Directions

InSAR cannot discern absolute volumetric changes in excavation pits and steep-gradient bench morphology; hence, multi-temporal LiDAR should be used in future investigations. InSAR provides better continuous spatial coverage for broad-scale subsidence, while LiDAR's discrete 3D point clouds give a topographic "ground truth" unaffected by active earth-moving phase decorrelation. This integration would decouple vertical settlement from horizontal mass wasting, providing a full picture of the mine's geomorphological development.

V. Conclusions

Field investigations and earlier studies in the Raniganj Coalfield have long indicated progressive GW stress associated with mining and urban growth; however, this study provides the first quantitative, long-term assessment of the magnitude and spatial structure of that depletion for the period 2002–2024. Using 17 integrated variables, including GRACE/GRACE-FO-derived GWS, the analysis reveals that major depletion zones ($>12.1 \text{ cm yr}^{-1}$) encompassed Durgapur, the Kataberia CBM fields, and the north-eastern mining areas, with maximum long-term decline rates reaching 13.4 cm yr^{-1} . The integration of 17 variables—including heat flux, canopy storage, and topographic indices—enables the model to proxy the complex water-energy balance and recharge-runoff dynamics inherent to the region's disturbed lithology. By accounting for these surface-subsurface interactions, the framework captures the nonlinear response of the coalfield's multi-aquifer system to both climatic variability and anthropogenic stress, ensuring the reconstructed trends are grounded in physical process dynamics. Pre-monsoon trends most effectively captured sustained aquifer exhaustion beyond seasonal variability. The strong spatial correspondence between GW withdrawal and SBAS-InSAR-derived land subsidence, with peak deformation rates of approximately $+21 \text{ mm yr}^{-1}$, demonstrates a direct coupling between subsurface water loss and surface instability. Meanwhile, fractal persistence ($H > 0.67$) confirms the long-term nature of the depletion. Additional well-based correlation analysis further supports the reliability of the reconstructed time series, with most stations exhibiting r values greater than 0.6. These results establish a quantitative linkage between GW depletion and land subsidence across the Raniganj coalfield. The findings provide critical evidence for strengthening GW regulation, mine-water management, urban planning, and mitigating subsidence risk. As mining expansion, urbanization, and climate variability continue to intensify GW stress worldwide, this study offers a robust scientific foundation for sustainable GW governance in data-scarce, mining-dominated regions.

Acknowledgment

We would like to express our sincere gratitude to the Central Groundwater Board of India for providing the GW well data. We would like to express our sincere gratitude to the European Space Agency (ESA) for its commitment to advancing scientific research and environmental monitoring, providing Sentinel-1 satellite data. PNSR (Corresponding author) acknowledges the ISRO Project (KCSTC-009/2025)/ IIT Kharagpur SRIC code (LDR).

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